



NYU | TANDON

Final Presentation

VIP –ML for Agricultural Commodity Market

TCN
Temporal
Convolutional
Network

**Final Alternative Data and
Preliminary Investment Strategy**

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Alternative Data for Wheat Futures (TCN Model) :Recap

1. Objective

Could Not Beat the Baseline TCN Model Result

- Enhance directional prediction of wheat futures by incorporating **cross-asset and macro alternative data** that capture global economic conditions, risk sentiment, and supply-demand dynamics.

2. Selected Alternative Data Features

- We have struggle to find one good datasets and explores some alternative way with the best model, TCN.
- We enhance wheat prediction by integrating cross-asset signals (oil, FX, volatility, crops) and compressing macro data into latent factors, enabling TCN to learn nonlinear market regimes.

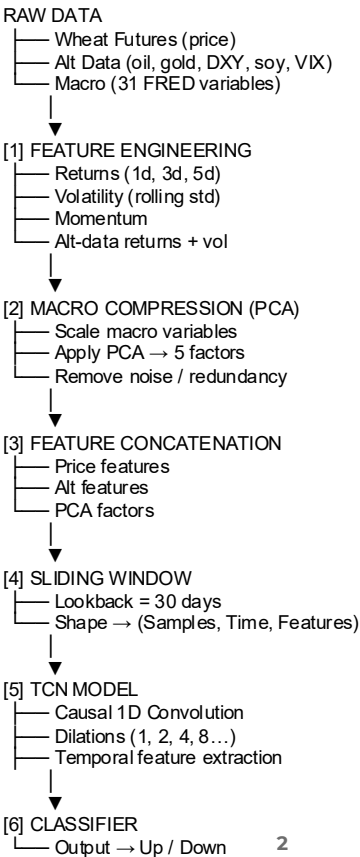
Macro Compression (FAVAR Approach) 31 macro variables → PCA (5 factors)

Category	Features
Price signals	returns, volatility, momentum
Alt data	oil, gold, DXY, soy, VIX
Macro	PCA factors (5)

- DXY: Dollar Index, VIX: Volatility Index, oil, gold and soy.

Still Tuning the model and methodology

- Validation metrics: **acc=0.5174 fl_macro=0.5168 roc_auc=0.5246**
- Validation predictions shape: (1034,)
- Baselines: majority acc=0.5358, fl=0.3489, auc=0.5000 | momentum acc=0.5019, fl=0.4993, auc=0.4993



1. Alternative Data Update: Narrower, Cleaner, More Honest

April Issue

- Broad Alternative-Data List was too Noisy for Daily Direction. **oil, gold, DXY, soy, VIX**
- Validation Looked Weak and Hard to Explain.
- Pipeline was Still being Tuned, so Claims were not Stable.

May Update

- Focused on Economically Motivated Daily Returns: Positive Pearson Cor **corn(.5752), soy(.3626)**
- Negative Pearson Cor **UUP (-.1174), CAD (-.1144)**.
- Combined 7 Wheat Features + 4 Cross-Asset Returns.
- Used Train-Fold-Only Scaling and PCA(5), then TCN.

Interpretation

- Small Improvement Over the TCN Baseline, but Not a Strong Signal.
- Larger Cross-Asset + Macro PCA Panel Underperformed on this Tail.
- The Update is better Framed as an Ablation, Not a Victory Lap.

Core Change: Move from 'Many Possible Alternative Data Sources' to a Leakage-Safe Comparison of Two PCA-augmented TCN tracks.

Wheat features + selected cross-asset returns -> fold-safe PCA -> TCN probabilities -> holdout evaluation

Pipeline	Acc.	ROC-AUC	FL_M
TCN Baseline Reference	0.541	0.526	0.541
TCN_PCA_WHEAT_ONLY: 7 wheat + corn/soy/UUP/CAD returns, PCA(5)	0.543	0.506	0.508
TCN_PCA: Yahoo overlay + FRED-MD + 4 returns -> macro PCA(5)	0.482	0.501	0.463

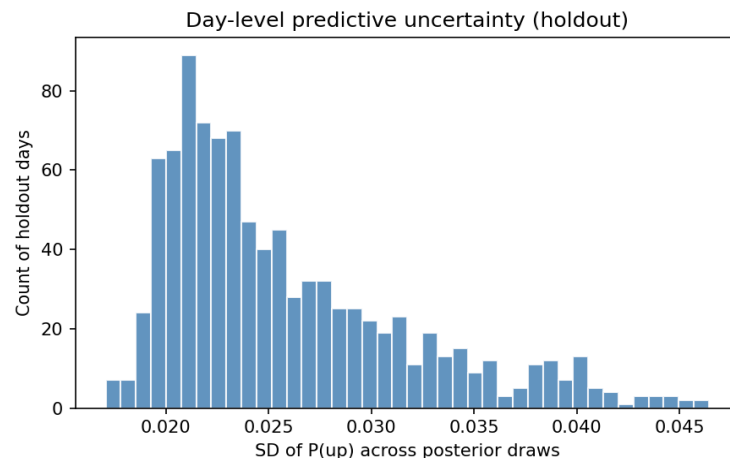
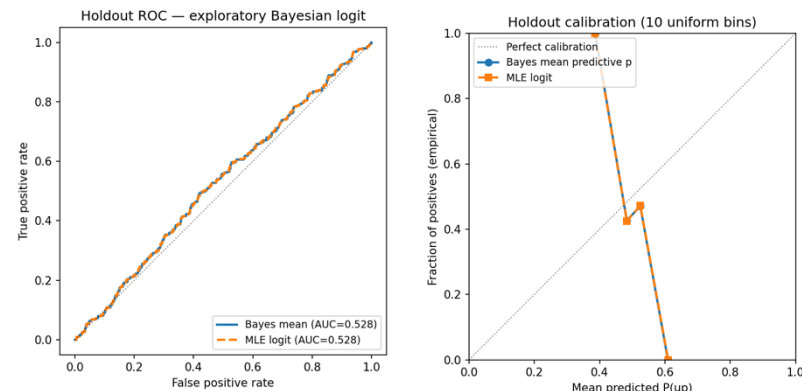
Replay: Adam LR=1e-6, 50 epochs, channels {128,64}, Apple MPS; terminal 15% chronological holdout; threshold 0.5 for Acc.

2. Strategy Needs Probabilities and Uncertainty

- ❖ Investment Overlay Consumes $P(\text{up})$, Not Just a Hard Up/Down Label.
- ❖ A TCN can Rank Days, but a Single Probability Does not Show Uncertainty.
- ❖ Bayesian Logistic MCMC(Markov Chain Monte Carlo) gives Posterior-Mean Probabilities and Dispersion.
- ❖ This Supplements the Strategy Layer; It Is Not the Champion Forecaster.

Model	Acc.	AUC	Brier
MLE Logit	0.500	0.509	0.255
Bayes Mean p	0.542	0.508	0.249

MCMC role: Expose Calibration and Uncertainty for
Threshold-Aware Long-Flat Rules.



Default design: same wheat + cross-asset PCA inputs as TCN_PCA_WHEAT_ONLY, flattened into a linear Bayesian logit.

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3. Investment Strategy: Tactical Long-Flat Holdout Proxy



- ❖ Retrain TCN_PCA_WHEAT_ONLY on Train + Validation Only.
- ❖ Score the Terminal 15% Holdout with Predicted P(up).
- ❖ Go Long Only When P(up) Clears a Hurdle; Otherwise Stay Flat.
- ❖ Proxy Only: No Commissions, Rolls, Margin, Options, Slippage, Storage or Live Order Timing.

Rule	Tot. ret.	Vol.	Sharpe	Max DD	% Long
Buy-and-hold	-0.468	0.319	-0.36	-0.613	1.00
P >= 0.52	0.000	0.000	--	0.000	0.00
P >= 0.50	0.009	0.005	0.51	0.000	0.001
P >= tau	0.000	0.217	0.11	-0.400	0.42
Momentum	-0.155	0.211	-0.11	-0.399	0.46

Interpretation: On One Bearish Holdout Tail, Sitting Out Part of the Decline Reduced Drawdown Versus Full Exposure. This is Risk Management Evidence, Not Proof of Alpha.

4. Discussion and Conclusion

What I Learned in this Semester ... and Conclusion

- ❖ Daily Wheat Direction Forecasting Remains A Weak-Signal Problem.
- ❖ TCN And PCA-Augmented TCN Produced The Best Descriptive Holdout Results, But The Improvement Was Modest.
- ❖ Chronological Discipline Matters More Than Model Complexity.
- ❖ Selective Alternative Data Helped More Than Broad Alternative Data Expansion.
- ❖ More Data Did Not Automatically Improve Performance.
- ❖ Leakage-Safe Design Was One Of The Most Important Lessons.
- ❖ Model Accuracy Alone Is Not Enough For An Investment Strategy.
- ❖ MCMC Helped Supplement The Strategy Layer With Posterior Probabilities And Uncertainty.
- ❖ The Long-Flat Strategy Reduced Exposure During A Bearish Holdout Window.
- ❖ The Strategy Result Should Be Interpreted As Risk Management, Not Proof Of Alpha.
- ❖ **Overall, The Project Became A Reproducible And Cautious ML Forecasting Evaluation.**

5. Future Work (Next Semester Tasks?)

1. Add **Alternative Trading Rules** Beyond Long-Flat, Such As **Long-Short, Volatility-Scaled Exposure, Confidence-Weighted Sizing, And Stop-Loss / Take-Profit Rules**.
2. **Extend The MCMC Model** So Posterior Uncertainty Affects Position Size, Not Only The Trade/No-Trade Threshold.
3. Compare TCN Probabilities, Bayesian Posterior-Mean Probabilities, **And Ensemble Probabilities** Under The Same Strategy Rules.
4. **Add Transaction-Aware Strategy Testing** With Commissions, Slippage, Bid-Ask Spread, Futures Rolls, Margin, And Contract Multipliers.
5. Run Walk-Forward Strategy Validation With **Expanding Windows** So Each Rule Is Tested Across Multiple Market Regimes.
6. Use MCMC Posterior Predictive Draws To **Stress-Test Strategy Outcomes**, Including Downside Risk And Probability Of Loss.
7. **Explore Regime-Aware Bayesian Models**, Where Parameters Or Thresholds Can Differ Across High-Volatility, Low-Volatility, And Macro-Stress Periods.



Thank You!

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